Lesson notes · for you

Machines that notice

IoT · Level 1 · Lesson 1

Time
45 min

Materials
Paper and a pencil for your table; Your eyes and a notebook on the way to school; An adult to ask when you're not sure what a device does

Learning goals

- Explain what a sensor does: it notices something and turns it into a number.
- Match your senses to sensors that do a similar job, and find senses machines have that you don't.
- Follow how a device senses, decides and acts.

Activity: Sensor hunt

1. Draw a table with four columns: the device, what it senses, what decides, and what it does. Leave room for at least eight rows.
2. At home, look for anything that switches, measures or warns by itself: a fridge, a solar lamp, a phone screen that dims, a smoke alarm.
3. At school and on the way, look at doors, taps, street lights, water tanks and buses. Write each one in your table.
4. Put a star by each sensor that notices something people can't: magnetism, very high sound, infrared, or a gas with no smell.
5. Choose your favourite find and explain its sense, decide and act steps to a friend or someone at home in under a minute.

Look, don't open: never take a device apart or touch sockets and mains wires. Ask before you photograph something that isn't yours.

Discussion

- Which of your senses would be hardest for a machine to copy, and why?
- When should a person, not a sensor, make the decision?
- What would you want a sensor in your town to warn people about?

Check yourself

- Name one device near you, and say what it senses, what decides and what it does.

Try it this week
Go on a sensor hunt: find eight things that switch by themselves, and write what each senses, what decides and what it does.

Worksheet

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Name _____ Class _____ Date _____

1 **Your school's water tank has a sensor that notices how full it is. When it reads low, the pump starts. Which part made the decision?**

- ◯ The sensor, because it noticed the water was low
- ◯ A small computer following the rule it was given
- ◯ The pump, because it's the part that moves

2 **Match each job to the sensor that does it.**

Turning a map to face north	• • • •	A light sensor
Warning a driver that a wall is close behind	• • • •	A carbon monoxide alarm
Taking a temperature without touching	• • • •	An infrared thermometer
Warning of a gas with no smell	• • • •	A parking sensor (sound too high to hear)
Switching a street light on at dusk	• • • •	A compass sensor (magnetism)

3 **A loudspeaker is a sensor, because it works with sound.**

- ◯ True
- ◯ False

4 **An automatic street light works through the night. Put its steps in order.**

- ☐ At dawn the reading rises, and the lamp switches off
- ☐ The lamp switches on
- ☐ The controller compares it with its rule: dark enough?
- ☐ The light sensor's reading drops
- ☐ Daylight fades in the evening

5 **Which of these sense something by themselves?**

- ☐ A hand dryer that starts when your hands are under it
- ☐ A torch you switch on
- ☐ A smoke alarm
- ☐ A light switch on the wall
- ☐ A fridge that keeps itself cool
- ☐ A loudspeaker

Which sensor around you do you rely on without noticing? What would happen if it stopped working?